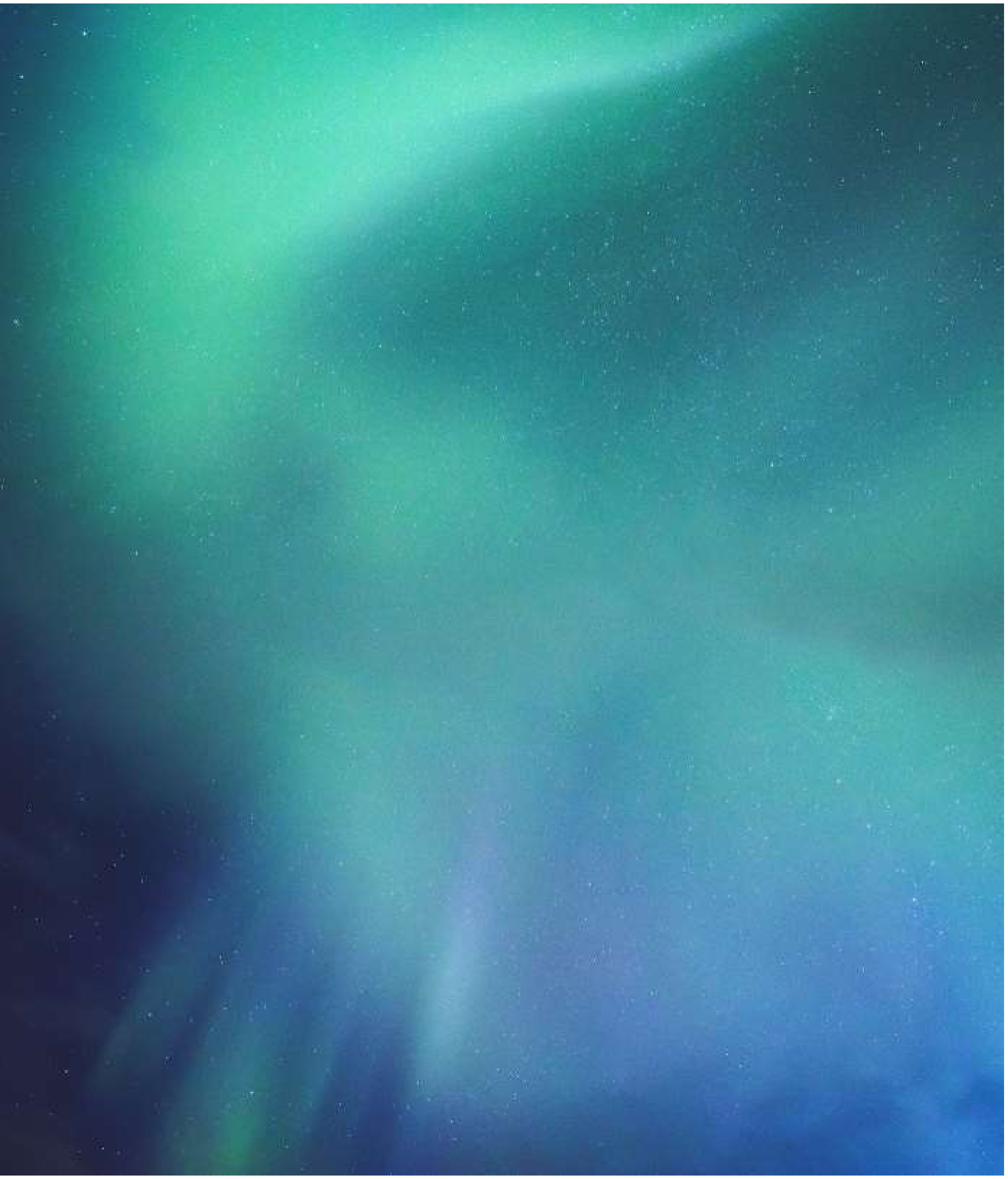
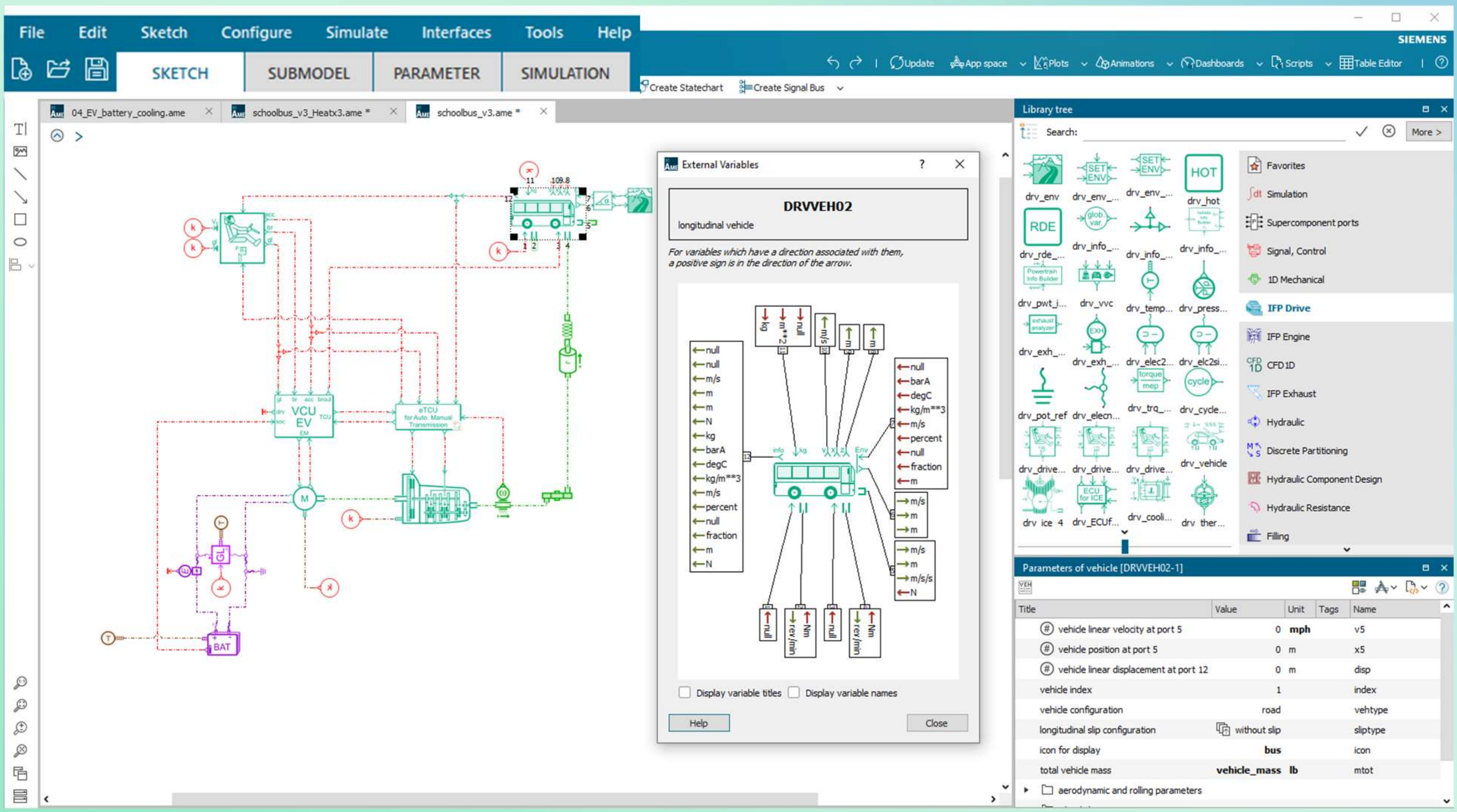


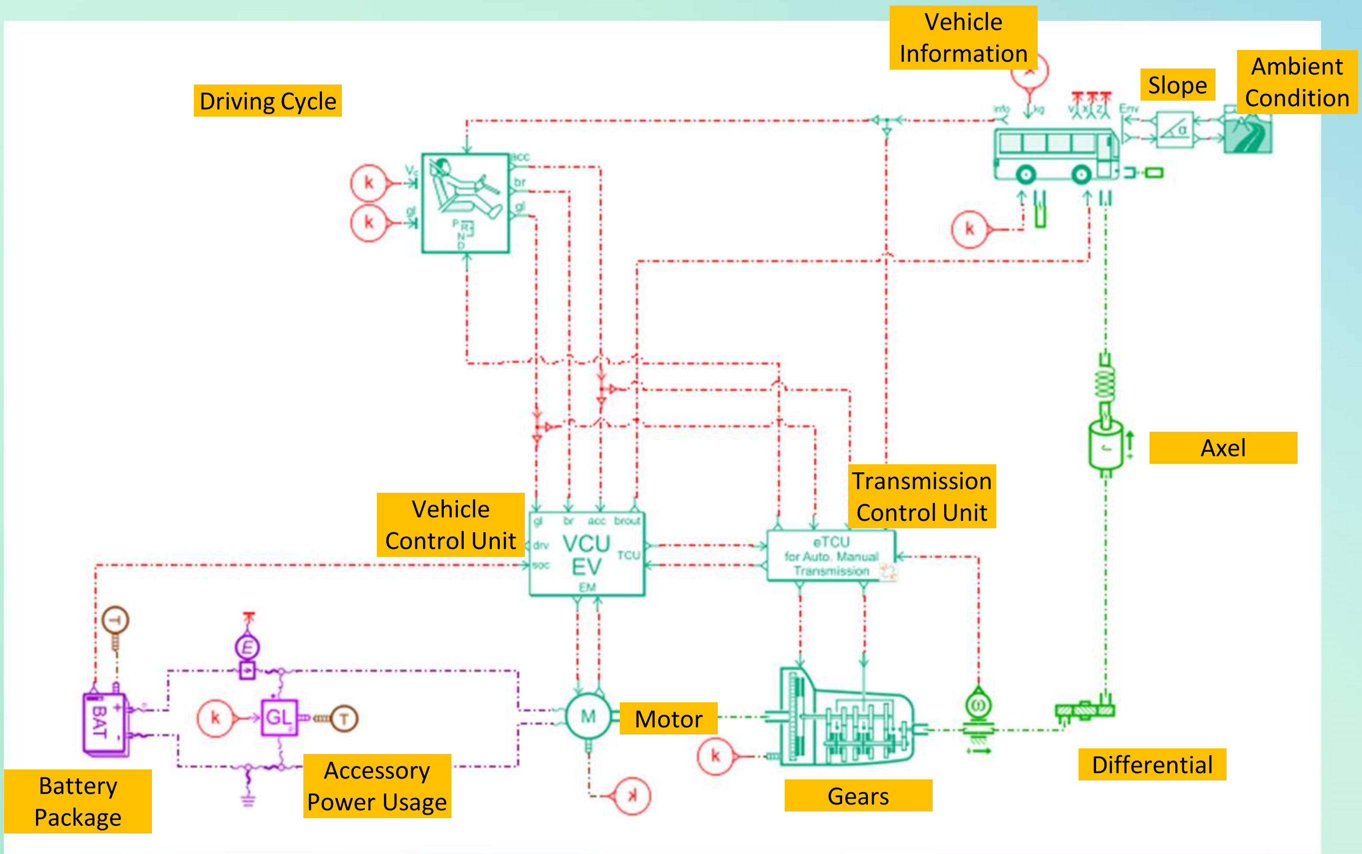
System Modeling of Battery Electric Vehicle

Dongmyung Suh

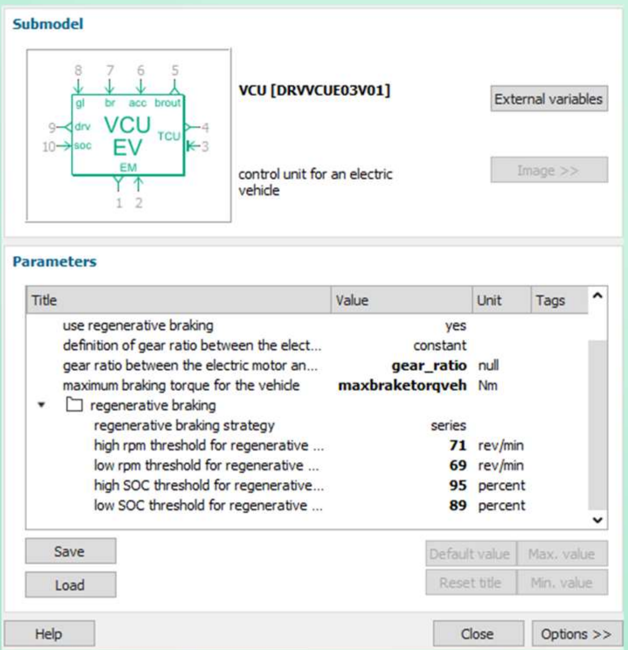
Amesim Model Development Environment



BEV Model

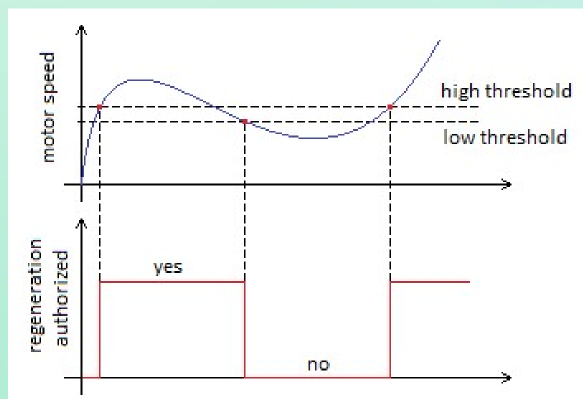

Vehicle Information
Driving Cycle
Slope
Ambient Condition
Axel
Vehicle Control Unit
VCU EV
Transmission Control Unit
eTCU for Auto. Manual Transmission
Motor
Differential
Gears
Battery Package
Accessory Power Usage
BAT
GL

Vehicle Control Unit (VCU)

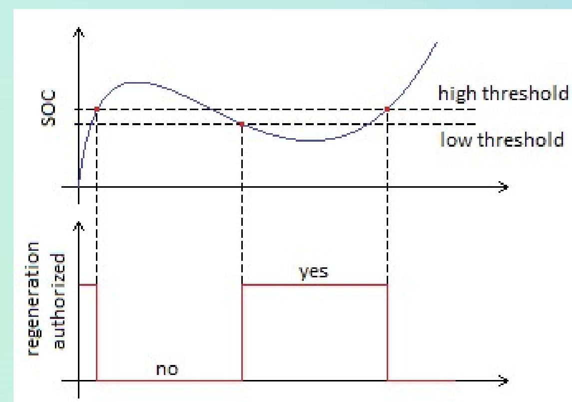


- Information from Driver, Battery and Motor
- Find the control to minimize the Battery power consumption
- Regenerative Breaking Control
- Send the control signal to Motor and Vehicle

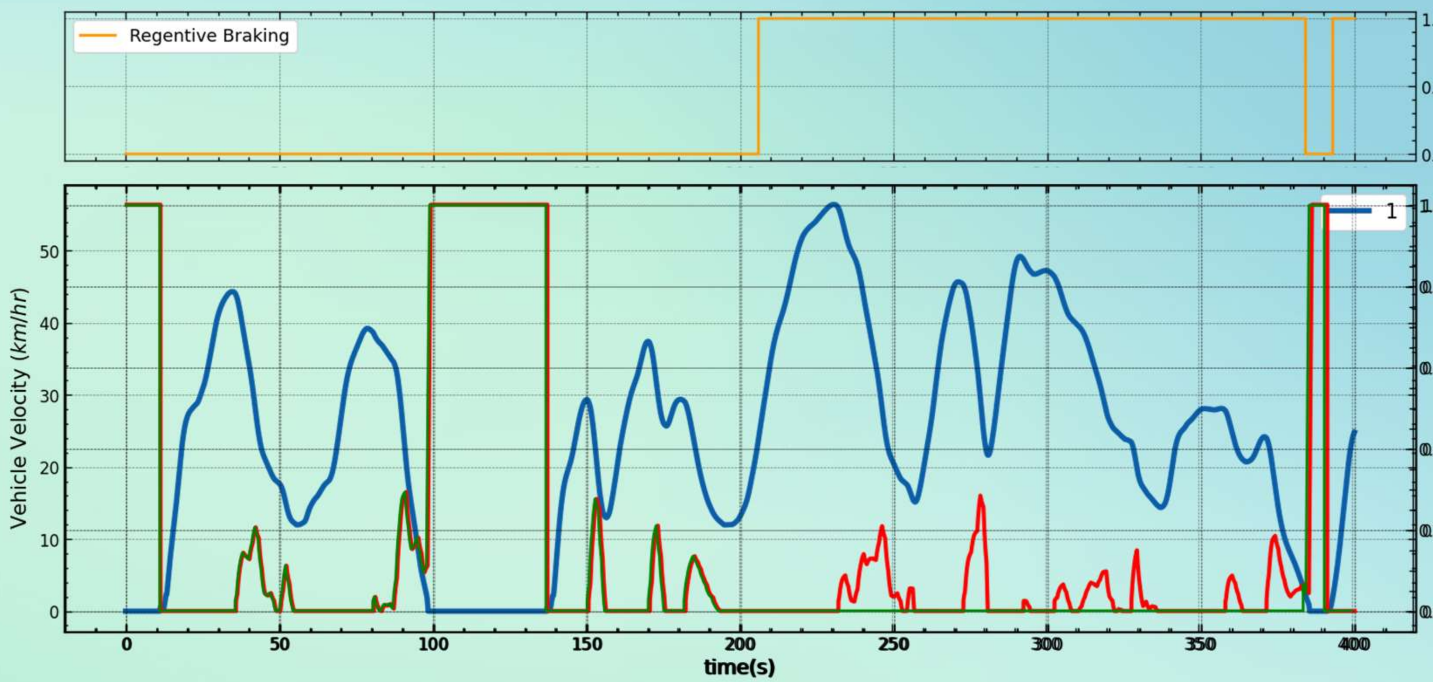
Thresholds on motor speed

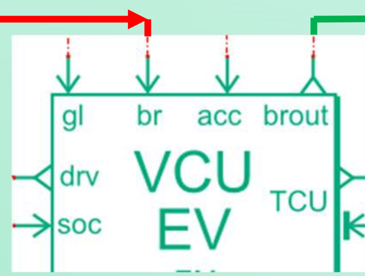


Thresholds on SOC

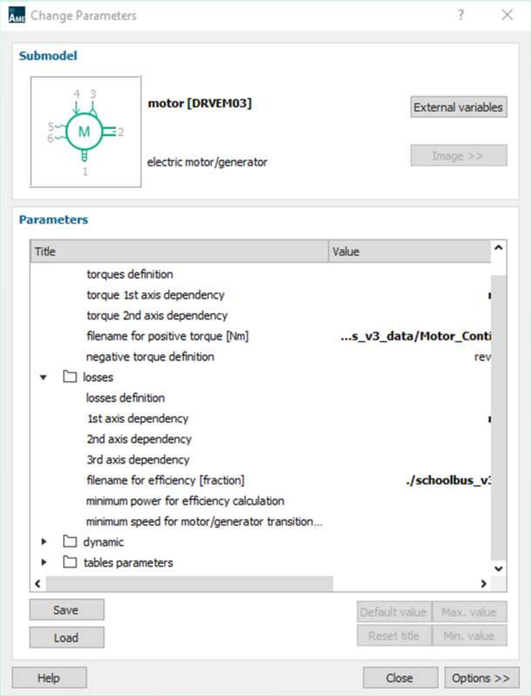


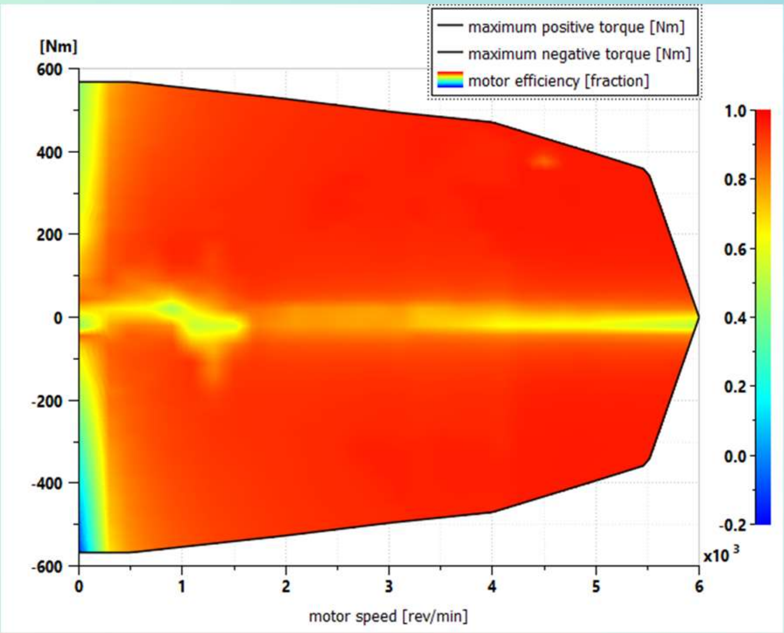
Accelera


Regentive Braking
Vehicle Velocity (km/hr)
time(s)
1


gl
br
acc
brout
drv
soc
VCU
EV
TCU

Motor/Generator





Battery Pack

- Equivalent Circuit Model / Electrochemical Model
- OCV & Resistance Data
- Entropic Coefficient : OCV change due to Temperature
- Diffusion and Charge transfer effects
- Aging effect / Thermal Runaway (not used yet)
- Battery electro-thermal identification Tool: Identify parameters from measurement data (not used yet)

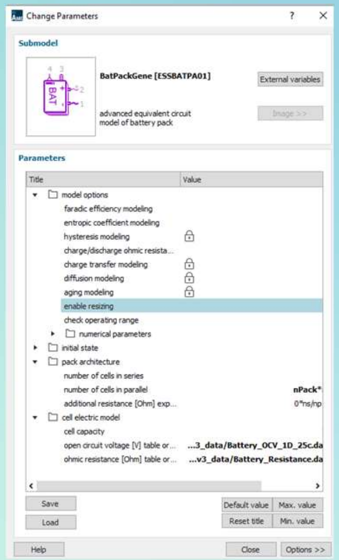
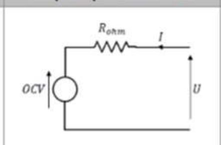
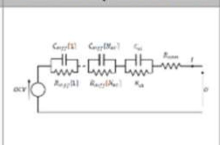
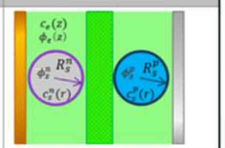


Table 2.2 Overview of the battery models

Modeling approach	Simple equivalent circuit	Advanced equivalent circuit	Electrochemical
Schema			
Thermal loss	Yes	Yes	Yes
Thermal runaway	No	Yes	No
Aging	No	Yes	Yes

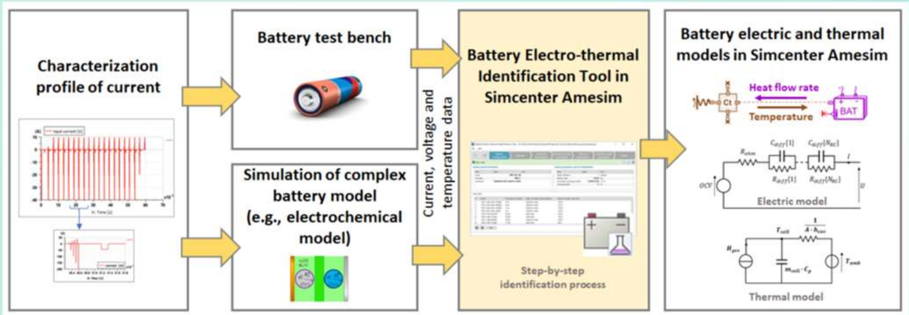


Figure 1: workflow of the Battery Electro-thermal Identification Tool

Product Information

BP97E Battery

Flexible Mounting | Higher Energy Density | Simple Infrastructure

With patented platform design and lighter than ever, BP97E packs one of the highest energy densities available today and is ideal for weight-sensitive applications. Flexible mounting options, including low-height sub-packs and in-chassis mounting, make integration easy for all applications.

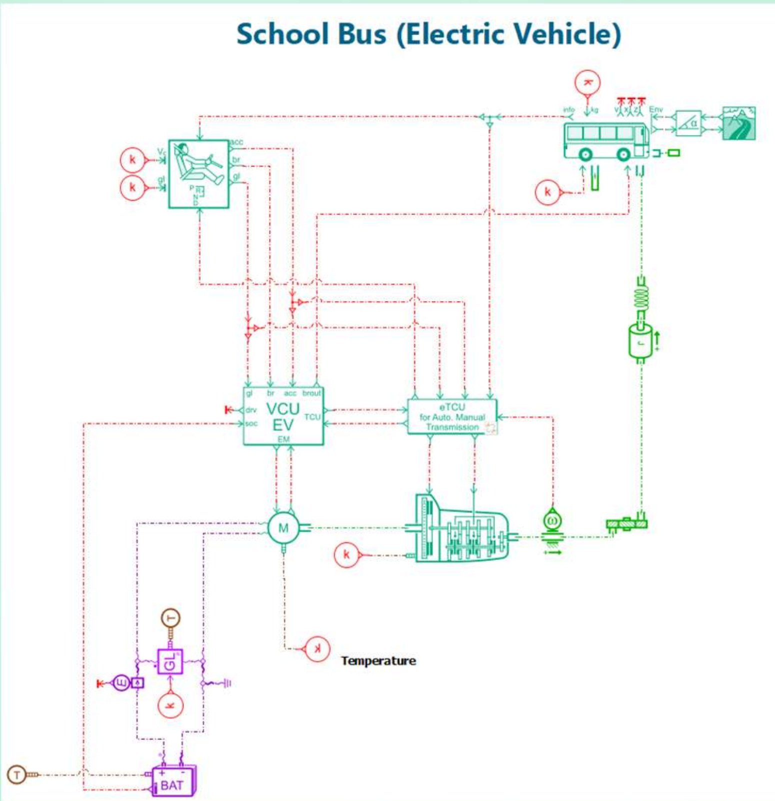
Product Specifications

Chemistry	Li Ion NMC-A	
Dimensions	1893 mm x 639 mm x 295 mm (excluding connections)	
Weight (dry)	550 kg	
Energy	Nominal: 97 kWh	Depth of discharge: 88%
Peak Power*	Charge: 209 kW (50% SOC)	Discharge: 131 kW (20% SOC)
Continuous Power	Charge: 82 kW (50% SOC)	Discharge: 105 kW (20% SOC)
Charging Capacity	0 - 80% DSOC in 1 hour	
Operating Voltage	583-689 V (618 V nominal)	
Cooling	H_2O and Ethylene Glycol, 50:50	
Ingress Protection	IP6K9K and IP67	
Cell Cycle Life**	> 4000 (C/3 charge/discharge, 25deg C, 80% DoD, 75% SOH)	
Operating Temperature Range	-40 to 50ºC	
Communication	SAE J1939	
ESS Configurations	Up to 8 in parallel	

*Peak power varies based on application

** Cycle life can be expected to change depending on use case, and is affected by operating temperature, power, and %DoD.

Amesim Model and Parameters



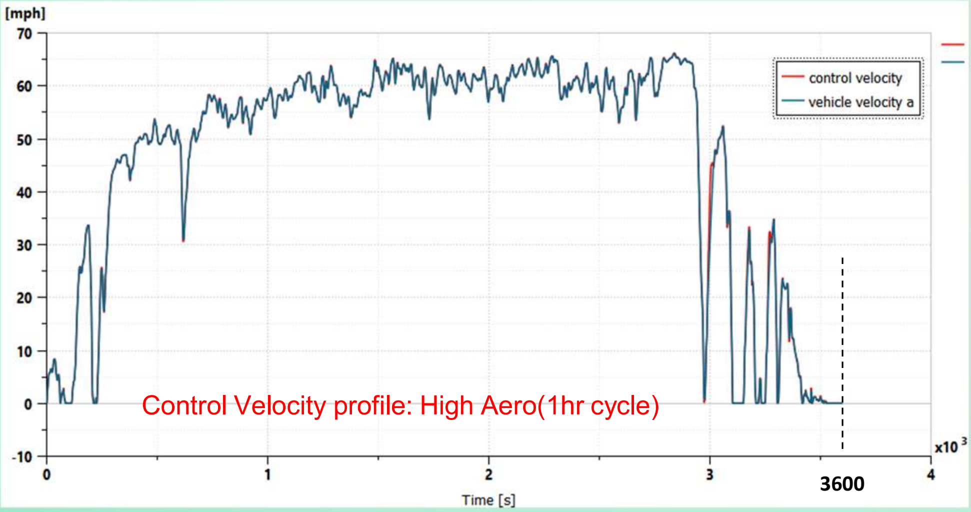
Np*ns*npack=2*168*2
1cell's capacity=78Ah
Total = 105k

Parameters

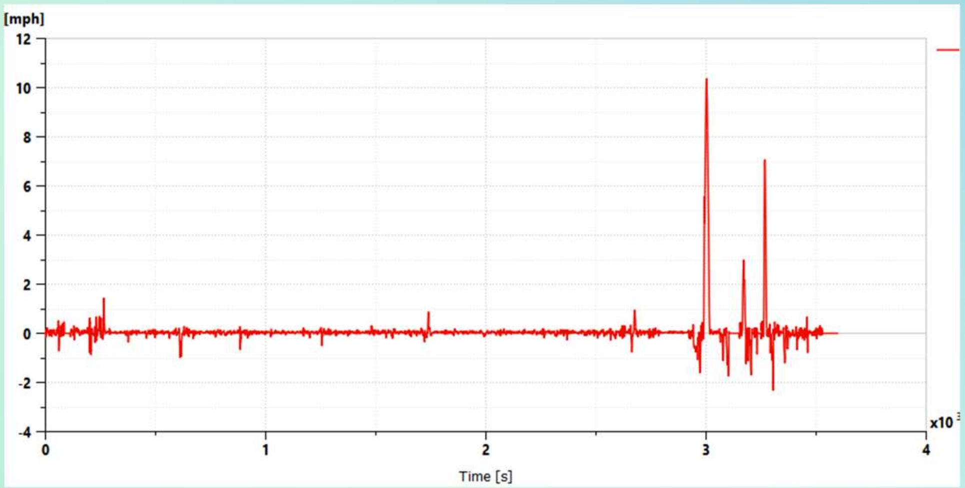
		Cyber	GT	
Model Setup / Allignment				
Vehicle	Mass	36200	36200	lb
	Cd	0.55	0.55	
	Frontal Area	7.43	7.43	m^2
	Crr	0.0061	0.0061	
	Tire rev/mi	497	497	
Motor	Name	V3p0_DanFossMotor_EffTestData	V3p0_DanFossMotor_EffTestData	
	Voltage	500 550 600 660 750	500 550 600 660 750	
	Peak Torque	567.8	567.8	Nm
	Cont Torque	567.8	567.8	Nm
Axle	Name	Axl_14Xe_2spd_3p25_hypoid_GT	Axl_14Xe_2spd_3p25_hypoid_GT	
	Ratio1	5.63	5.63	
	Ratio2	2.79	2.79	
	Hypoid	3.25	3.25	
Battery	Name	BP97E_2Pack	BP97E_2Pack	
	Source	Nexus Power Map EOL-School bus Base - 30012024.xlsx	Nexus Power Map EOL-School bus Base - 30012024.xlsx	
	# of Packs	2	2	
	Cells is Series	168	168	
	Cells in Parallel	2	2	
Accessories	ElecAccLoad_Stationary	5500 W	-	
	ElecAccLoad_Moving	6000 W	-	
	DCDC Energy	-	Different values for each duty cycle.	
	EPAS Energy	-		
	Air Compressor Energy	-		
	BTMS Refrigerant Compressor Energy	-		
	BTMS eHeater Energy	-	Resolves during Runtime	
	Cabin Refrigerant Compressor Energy	-		
	Cabin eHeater Energy	-		
Motor Brake Controller	C_MCAH_Base_DL_SpdTrq_Table	Need to update on CyberApps		
	C_MCAH_Base_DL_SpdTrq_Brk_Table			
	C_MCAH_Base_DH_SpdTrq_Table			

Accelera

Simulation Results



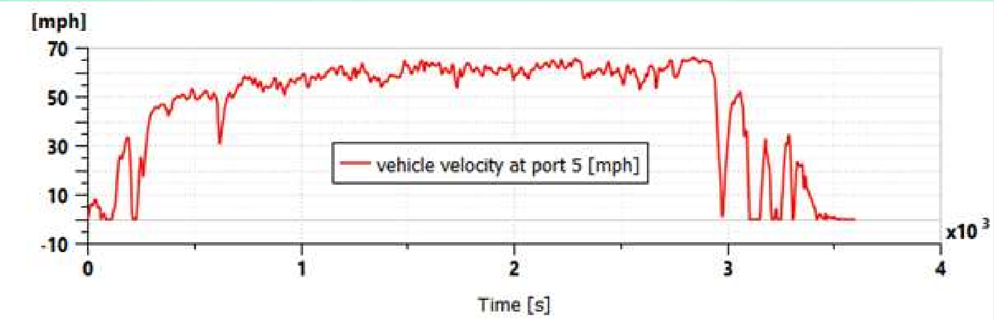
Velocity

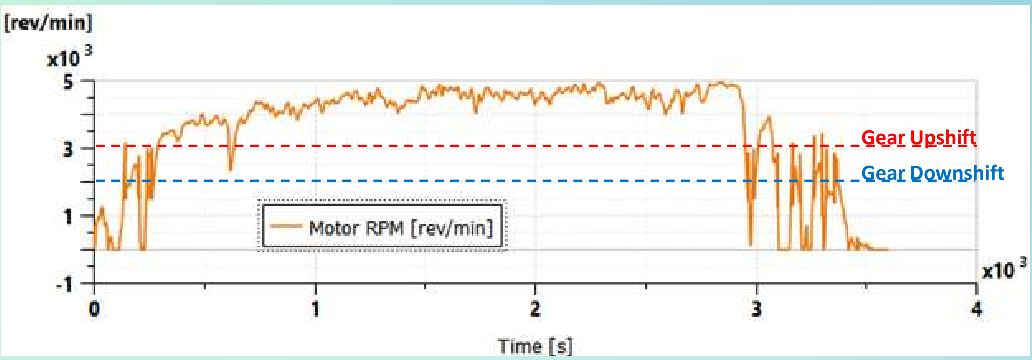


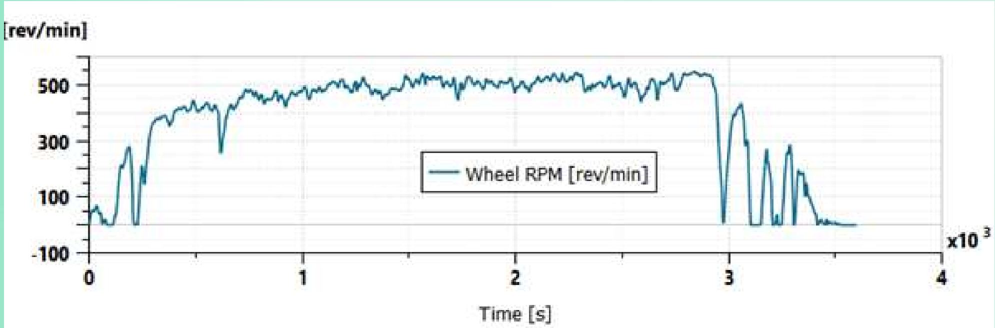
Velocity Difference

Accelera

Velocity/Wheel/Motor/Gears

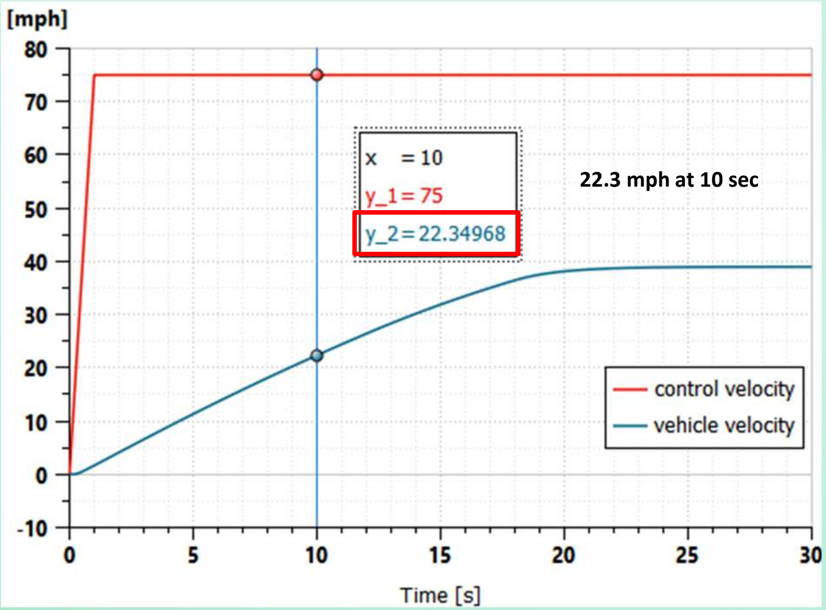




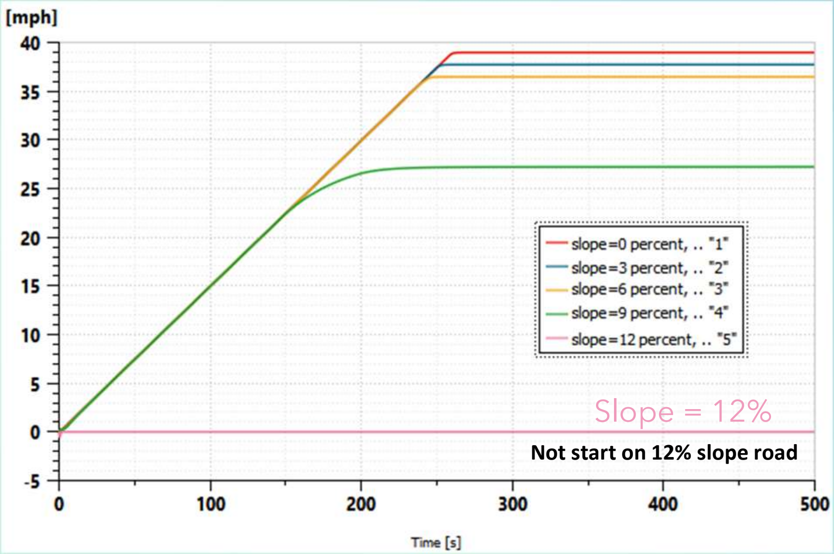


Accelera

Vehicle Acceleration, Slope Start



Vehicle Acceleration

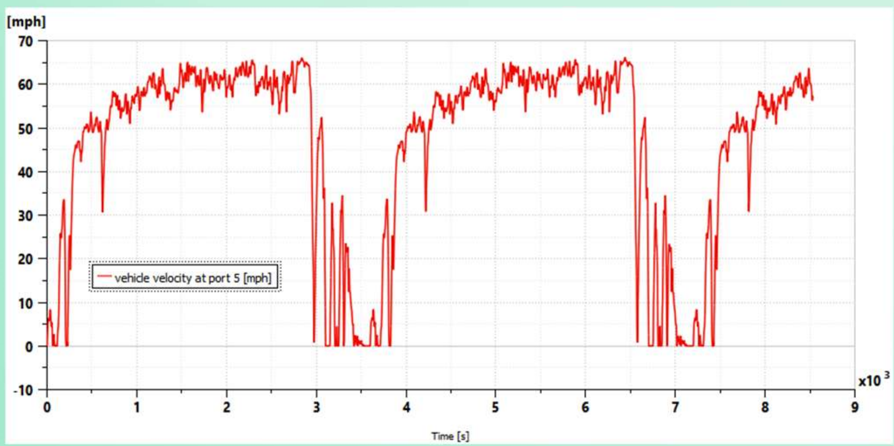


Vehicle Start at slopes

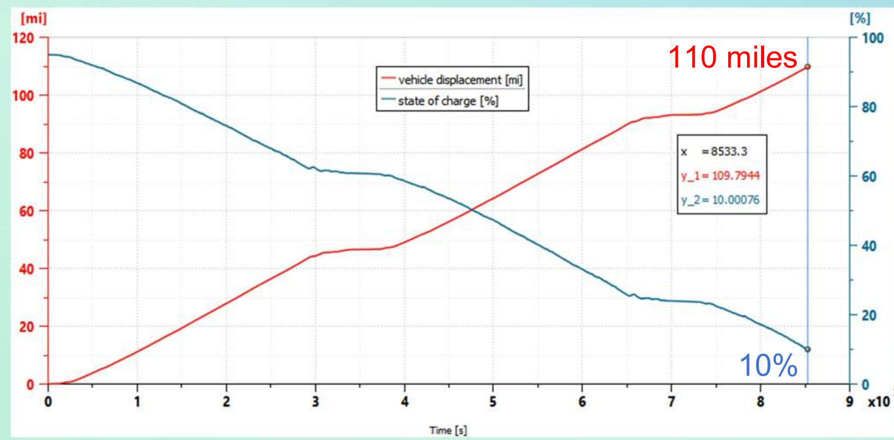
Accelera

Driving Range

- How far can the bus drive while SOC changes 95% to 10%?
- Driving Cycle: High Aero



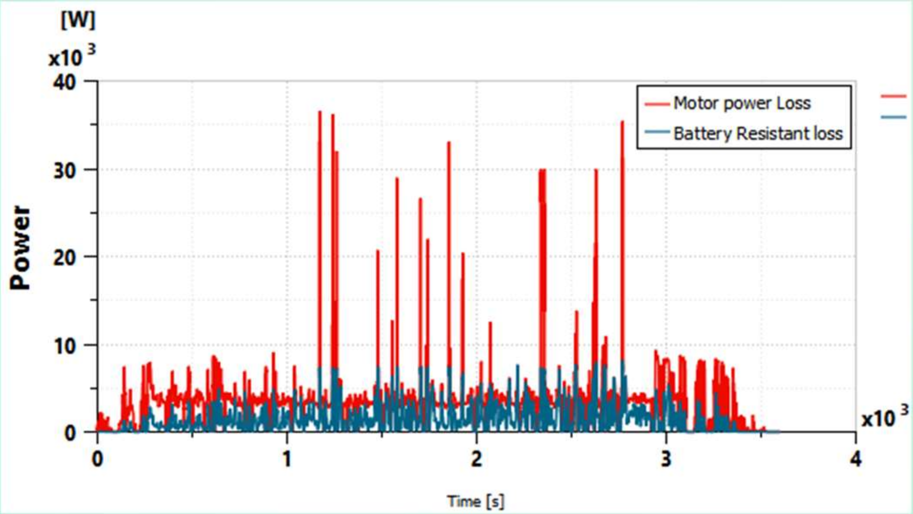
Velocity



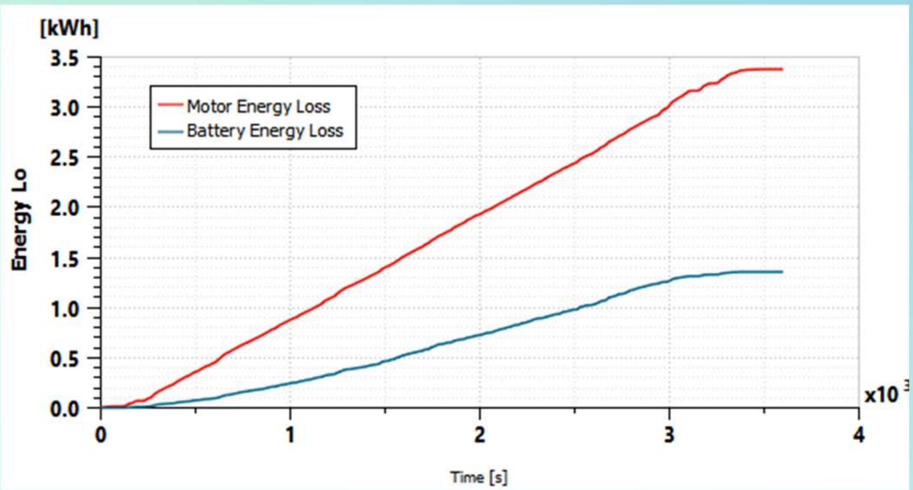
Distance & SOC

Accelera

Heat Loss



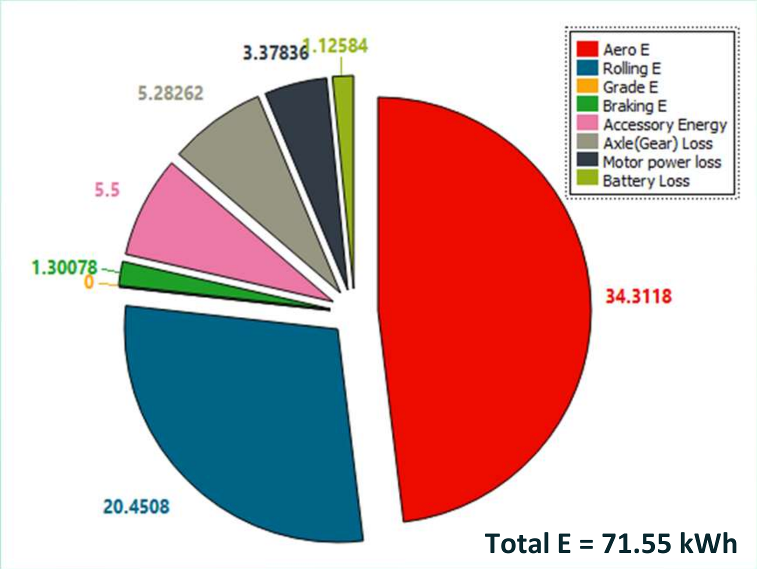
Power Loss [W]



Energy Loss [kWh]

Accelera

Energy Usage



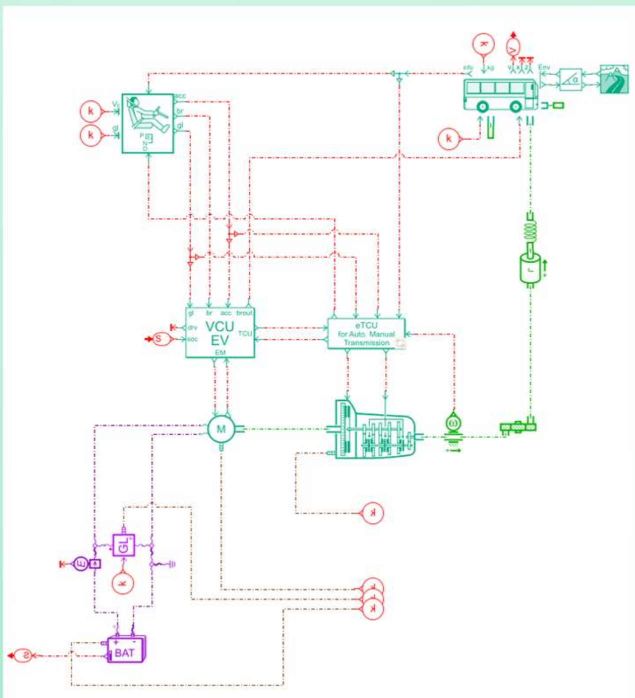
Energy Consumption (kWh) after Driving Cycle

(Unit: kWh)

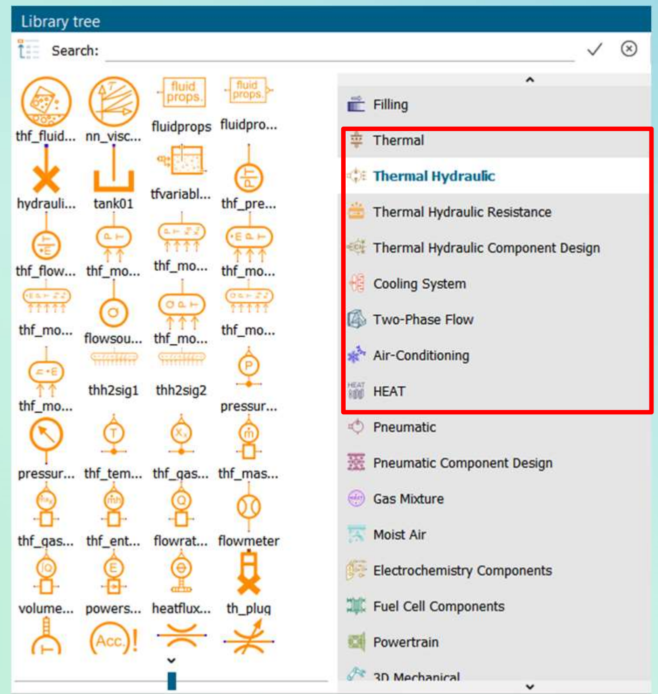
	Cyber	GT	Amesim
End SOC %	48.50	47.95	**55.5**
Battery Electric E	**86.74**	**84.53**	**70.21**
Aero_E	33.18	32.60	34.3
Roll_E	20.38	20.49	20.45
Grade_E	0.00	0.00	0
Brake_E	1.77	1.83	1.30
DCDC Energy	-	0.94	-
EPAS Energy	-	1.93	-
Air Comp_E	-	17.13	-
Accessory_E	19.69	-	5.5
Axle_E	5.61	5.61	5.28
MG1_E	3.17	3.28	3.34
Battery Heat_E	2.94	1.95	1.36
Total Energy	86.74	85.76	71.59

Accelera

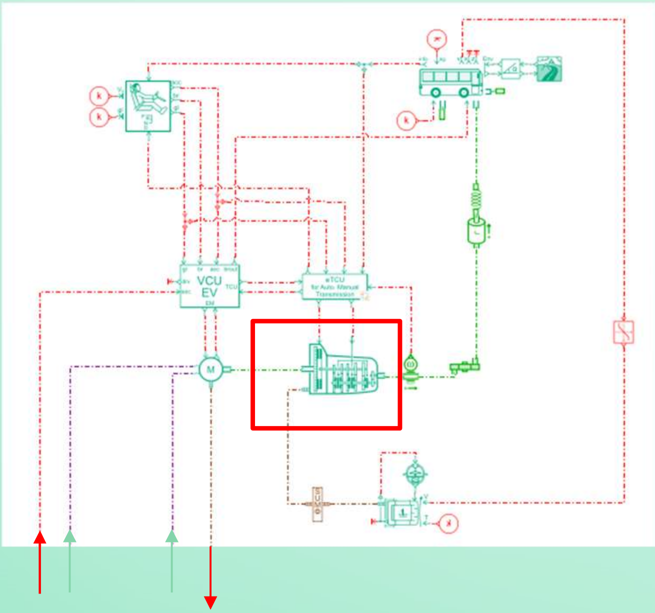
Excretion of Heat

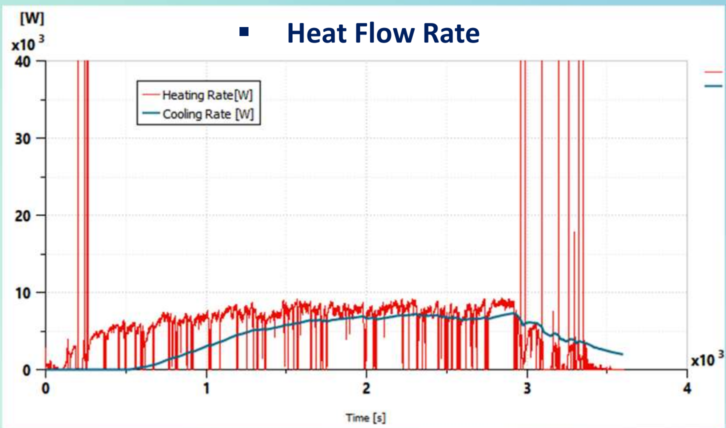


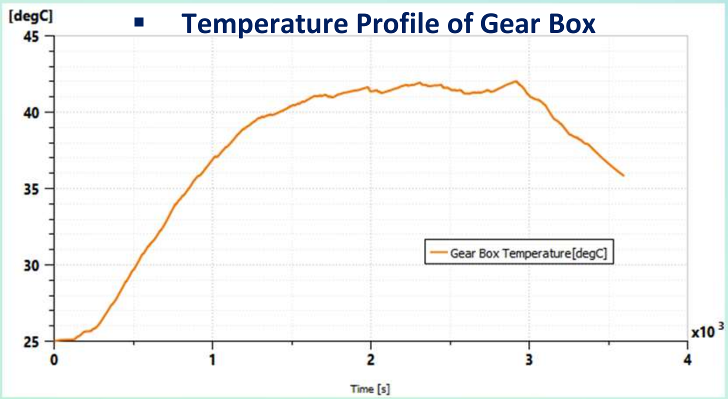
Library of Thermal Analysis



Gear Box

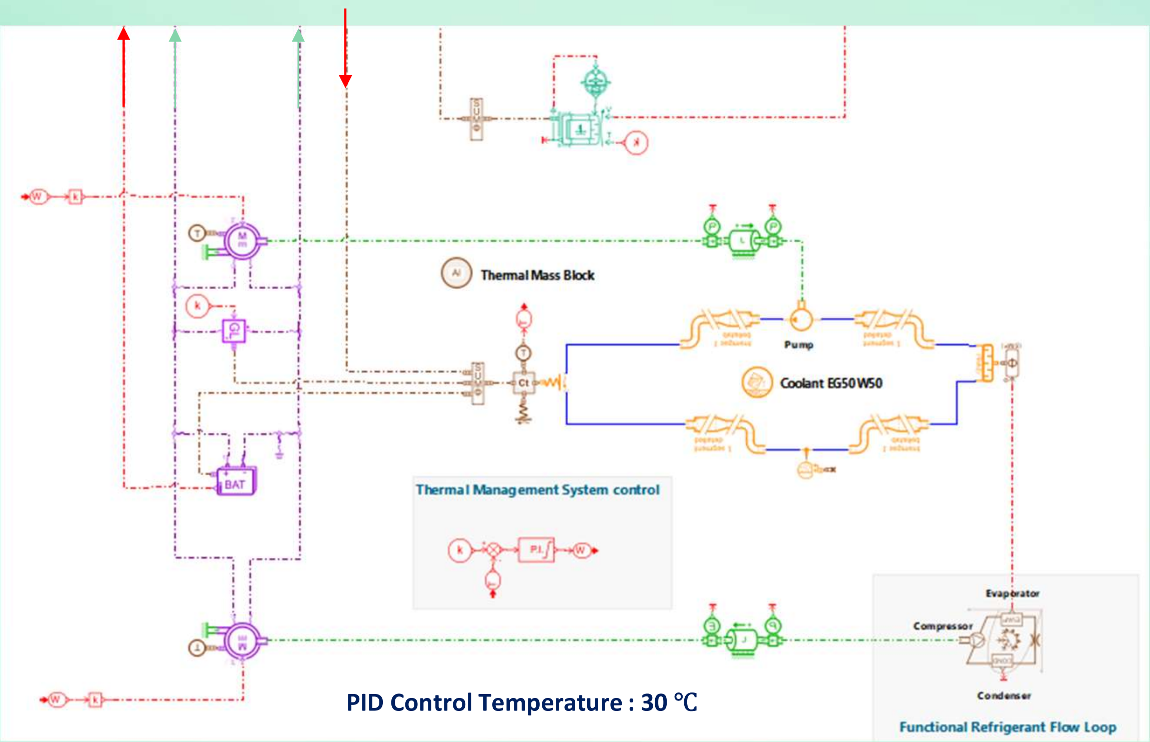

VCU
EV
eTCU
for Auto Manual Transmission
M


[W]
x10 3
Heat Flow Rate
Heating Rate[W]
Cooling Rate [W]
40
30
20
10
0
x10 3
0
1
2
3
4
Time [s]

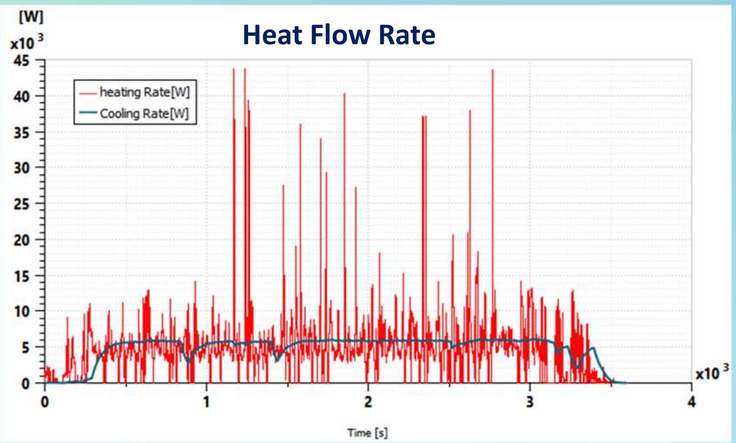

[degC]
Temperature Profile of Gear Box
45
40
35
30
25
Gear Box Temperature[degC]
x10 3
0
1
2
3
4
Time [s]

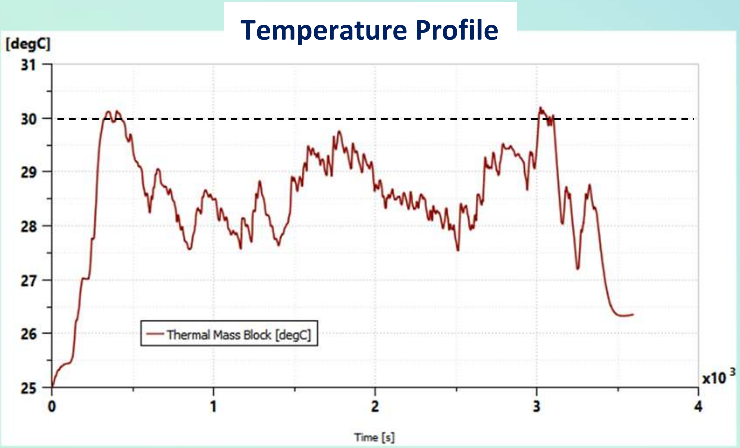
Accelera

Motor / Battery / Accessory



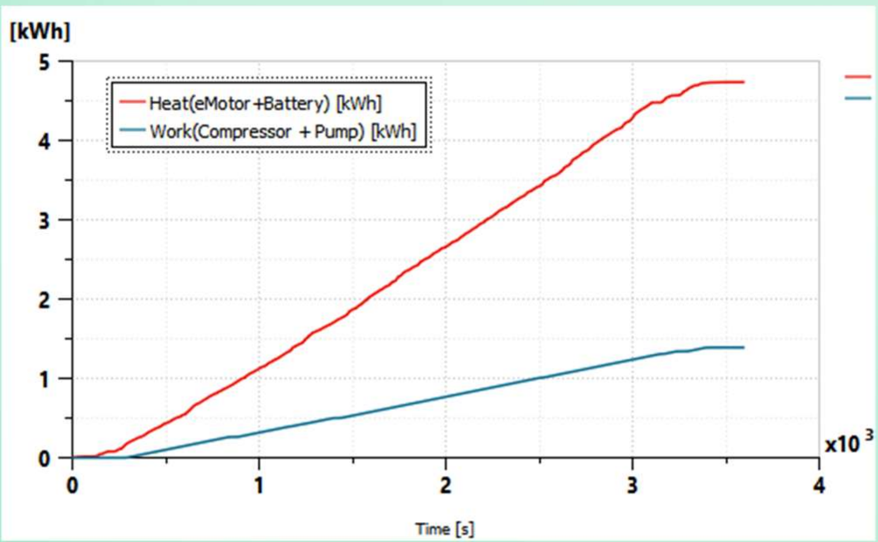
PID Control Temperature : 30 ℃

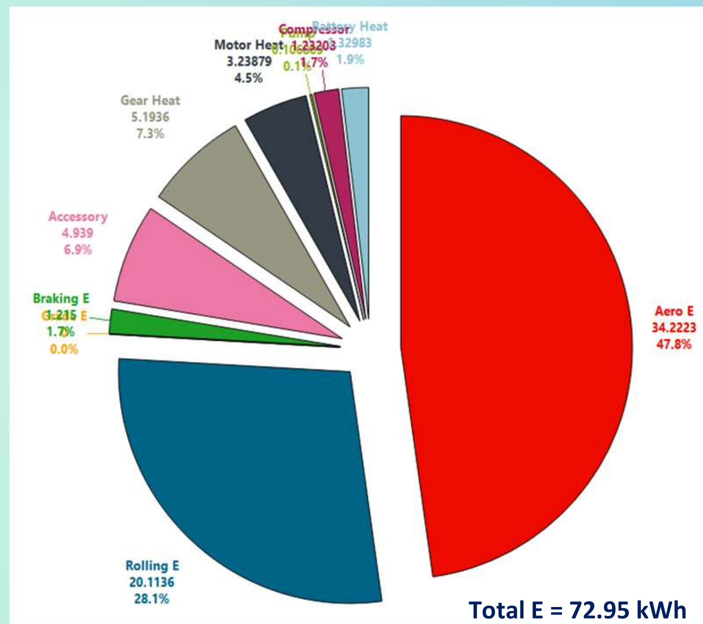




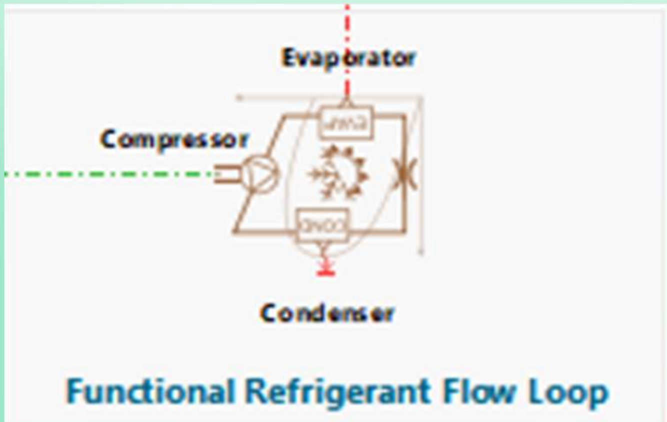
Accelera

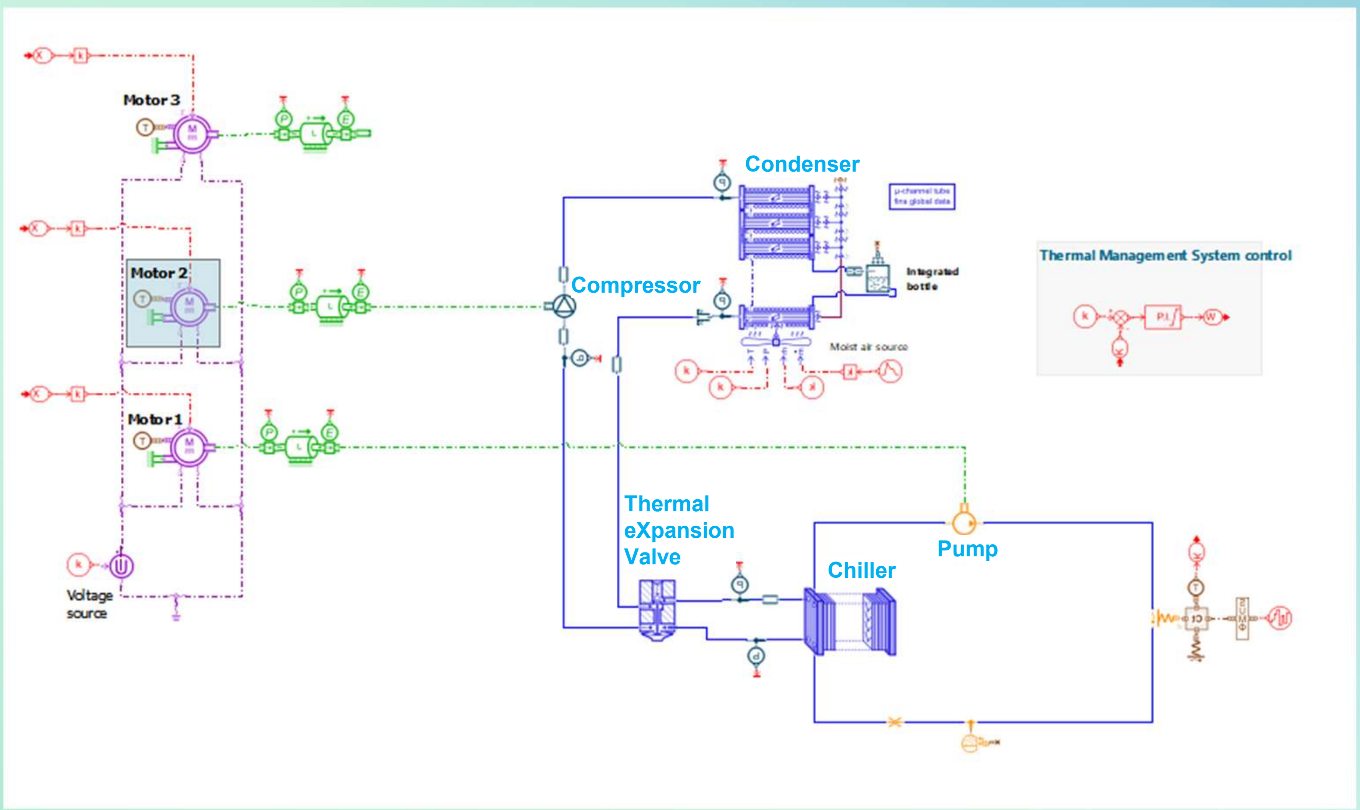
Energy Analysis





Detailed Model of Refrigerant Cooling Cycle





Conclusions

- ❑ A BEV System model is developed using Amesim.
- ❑ It can be used to calculated a Vehicle diving performance (Velocity, Wheel/Motor RPM, Gear, Battery Consumption, Mileages).
- ❑It can also be used for Thermal Manage System modeling.
- ❑ The model can be easily extended for considering more detailed and comprehensive physics.